Multidimensional Goodhart

How controlling for a proxy reshapes, rather than removes, its error

Working draft — Chapters 1–3

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1 Goodhart's law, multidimensionally

1.1 The shape of the problem

Goodhart's law is usually quoted in one of two forms. The popular slogan, which is actually Strathern's compression [11]:

> When a measure becomes a target, it ceases to be a good measure.

and Goodhart's own statement [3]:

> Any observed statistical regularity will tend to collapse once pressure is placed upon it for control purposes.

Both are stated about a **scalar** measure: a single number that was correlated with something we cared about, until we started optimising it. The thesis of this book is that the scalar framing hides most of the structure. Real goals — the goals of a person, a team, a company, a state, or a trained model — are **multidimensional**, and the proxies we use to steer toward them are multidimensional too, usually with a different (and smaller) set of dimensions. Once that is taken seriously, "the regularity collapses" is replaced by a more precise and more useful question: **how does pressure on a proxy reshape the distribution of its residual error?**

The answer depends on the channel. In a **selection** regime, the policy reweights a fixed baseline distribution; hidden drift is governed by the baseline response of hidden coordinates to the selected proxy, and dimensionality matters only through coupling and variance structure. In an **intervention** regime, agents respond; probability mass moves to states the baseline did not contain, and the drift is governed by cost geometry, stakes, and available gaming channels. A broader recursive-Goodhart hypothesis is that repeated proxy refinement often pushes the remaining error into dimensions that are less legible to the evaluator. This book treats that as a hypothesis suggested by the framework, not as a theorem of the framework.

So there are three layers to keep separate. First, the formal chapters prove or derive toy results about selection response, baseline drift bounds, intervention channels, and multidimensional gaming. Second, those models imply conditional design lessons: additive scorecards, conjunctive scorecards, and different harm-per-score exchange rates behave differently. Third, the recursive story — that residual error may migrate into less-monitored or harder-to-elicit dimensions as the proxy stack is patched — is an empirical conjecture. The job of these first chapters is to make all three layers statable without pretending they have the same epistemic status.

Remark. **Licensed claims.** These chapters license three narrow uses. First, when pressure only reweights a fixed baseline distribution, the right quantities are selection-response functionals of that baseline: covariance locally, threshold response in the tails, and a chi-square drift budget for bounded selection. Second, when agents can change the state-generating process, the drift budget is not a baseline statistic; in the quadratic Stackelberg toy model it is $\sqrt{2\kappa V}$, a function of gaming ease and stakes. Third, for intervention channels with compensatory additive scores and separable quadratic gaming costs, adding an independently gameable measured dimension increases aggregate gaming capacity K_M, lowers the minimum cost of clearing a fixed score gap, and weakly expands the population of agents for whom gaming pays. These claims are meant for designing scorecards, benchmarks, and evaluation suites; they do not prove the recursive-Goodhart hypothesis or generalize automatically to non-convex training dynamics.

The starting point is the variant taxonomy of Manheim and Garrabrant [6] — regressional, extremal, causal, and adversarial Goodhart — and Wentworth's geometric reconstruction of the regressional case [12], plus Wentworth's later warning that experiments and metrics often measure a different latent quantity than their designers think they do [13]. The economic ancestor is the multitask principal-agent line: Holmstrom and Milgrom's canonical model of measurable and unmeasurable tasks [5], Baker's performance-measurement model [1], and Prendergast's survey of incentive provision [9]. In AI safety, the closest formal neighbours are reward gaming and misspecification models [7, 10] and the partial-objective model of misaligned AI [14]. The contribution here is to put these literatures in one frame: the Goodhart taxonomy supplies the failure modes, strategic classification and performative prediction supply adaptive distribution shift, principal-agent theory supplies substitution toward measured tasks, and the selection/intervention split makes the vector structure of both regimes explicit.

1.2 A worked intuition: the hierarchy of proxies

Consider a company with leadership whose real objectives are something like *make a product customers want*, *make money*, *enjoy the work*, *gain prestige*, and a dozen other things, in some weighting that no written document captures. Call that vector of objectives G. Each quarter the leadership sets a proxy P: a handful of metrics — revenue, delivery dates, an NPS number. The teams that implement P have their own objectives, and the proxy they **actually** optimise, P_i, folds their incentives into the official one. Its dimensionality is at least that of the most complex implementer's goals.

Now suppose leadership notices that Team 1 is "optimising too hard on the metrics" — the dashboard looks better than the product. They intervene. The correction they apply is *not* the gap between G and Team 1′s true output. It is the gap between the **felt proxy** by which leadership noticed the problem and Team 1′s measured performance. The error term is being filtered through several layers of proxy, each with its own residual.

Two things follow, and they recur throughout the book:

1. When a principal constrains an agent whose behaviour has more degrees of freedom than the principal's proxy, the proxy must compress, ignore, or invent coordinates. Some residuals are ordinary measurement errors. Others are closer to incomputable noise: not noise the principal is failing to measure well, but behaviour whose relation to the principal's goal has not been represented in the proxy space at all. This is the **dimension gap**: if the proxy map has no coordinate for a goal-relevant degree of freedom, optimisation can move that degree of freedom without registering as proxy error. The later formal claims are about special cases where that movement is observable as either selection response or intervention cost.
2. Whether the multi-layer correction process converges depends on whether the layers' perception errors are roughly independent. If each layer's residual is its own, a dimension of error that one layer sees can be corrected even though no layer sees the whole picture. If the residuals are strongly correlated — a centrally-planned organisation, a single dominant metric — the system is fragile: every layer is blind in the same direction at once.

Remark. There is an ML reading of all this, but it is diagnostic rather than a finished model of training. Pretraining, validation-set selection, and model selection are selection regimes when they choose among behaviours already present in the candidate distribution. RLHF and reward-model optimisation become intervention-like once the learner changes its policy to exploit the reward channel; this is the reward-gaming setting studied by [10] and [7]. What plays the role of κ in a neural network is not settled: it might be gradient accessibility, representation density from pretraining, benchmark contamination, reward-model feature simplicity, or optimiser search efficiency. The framework below therefore does not claim that quadratic Stackelberg gaming describes RLHF. It says what an ML version would have to estimate: baseline response curves for selection; cost geometry, stakes, and available gaming channels for intervention.

1.3 Setup: goals, proxies, and two kinds of gap

Fix a state space S — the set of configurations the world (or the organisation, or the model) can be in. A **true goal map** is a function $G : S \to \mathbb{R}^m$; its m coordinates are substantively distinct things the principal cares about. A **proxy map** is $P : S \to \mathbb{R}^k$, the k things actually measured and acted on. The principal also has in mind an **intended model** $\varphi : \mathbb{R}^m \to \mathbb{R}^k$ — "the proxy should be roughly φ of the goal" — and the discrepancy is the **residual**

$$\varepsilon(s) = P(s) - \varphi(G(s)).$$

This little decomposition already separates two distinct Goodhart channels:

- A **dimension gap**: $\ker \varphi \neq \{0\}$, so some directions in goal space are invisible to the proxy. No amount of measuring P accurately tells you anything about movement along $\ker \varphi$.
- An **observation gap**: $\varepsilon \neq 0$, so even the directions the proxy is meant to track are corrupted — by noise, by lag, or (Chapter 3) by deliberate manipulation.

Toy example. Let product quality be $G = (\text{reliability}, \text{delight})$ and let the proxy be uptime. Then "delight" lies in $\ker \varphi$ — the dimension gap. Noisy uptime instrumentation is part of ε — the observation gap. A research-evaluation example: citation count is **intended** to track research quality; "long-run conceptual fertility" may be in $\ker \varphi$, while bot citations and citation cartels are residual variation in proxy space.

Remark. A caution that will shape the notation. If the principal's "true" objective is genuinely scalar, then $\ker \varphi \neq \{0\}$ may be an artefact of having written that scalar goal in redundant coordinates. So dimension-gap claims should always be made **after** choosing coordinates whose components are substantively distinct goal variations, not arbitrary embeddings. We will keep $\varphi : \mathbb{R}^m \to \mathbb{R}^k$ as the principal's intended correspondence and consistently distinguish three subspaces:

- $\ker \varphi$: goal variation invisible to the intended proxy;
- $\operatorname{im} \varphi$: proxy variation intended to track goal variation;
- residual proxy directions: variation of P not explained by $\varphi(G)$.

1.4 A first specialisation, and a roadmap

Throughout Chapter 2 we work the simplest non-trivial case: $G(s) = X \in \mathbb{R}^m$ with X Gaussian, a single scalar proxy that is one coordinate (or one linear combination) of X plus noise, and **selection** — keep the states scoring above a threshold. This isolates the two classical "easy" Goodhart effects, **regressional** and **extremal** Goodhart [6], from the harder ones. It is the right first probe precisely because it is the regime where, as Chapter 2 shows, hidden harm turns out to be **bounded by quantities visible in the pre-selection distribution**. Chapter 3 then asks what changes when the principal's policy does not merely re-select from a fixed population but changes the population itself — when agents **respond** — and shows that this is where the "deep Goodhart" story has teeth, and where the bound from Chapter 2 has no analogue.

A reader who wants the punchline before the construction: there is a clean dichotomy between **selection channels** (the policy reweights a fixed baseline distribution; all classical regressional/extremal Goodhart lives here; hidden drift $\leq \delta \cdot \|s\|$ with every term a baseline functional) and **intervention channels** (the policy moves probability mass to where the baseline had none; causal and adversarial Goodhart live here; no baseline bound exists, and any bound must be imported from a model of what the responding agents can afford to do). Chapters 2 and 3 are those two halves.

The appendices are organised the same way. Appendices C–F are visual aids for claims made in the formal chapters: selection response, dimensional coupling, selection versus intervention, additive versus conjunctive gaming, and the exchange-rate condition for conservation. Appendix G is deliberately different: it is a speculative cartoon of the recursive-Goodhart intuition. Its purpose is to say what the framework might help test, not to smuggle an additional theorem into the paper.

1.5 Summary of results

The proved or directly derived results in these chapters are:

- Gaussian threshold selection shifts hidden means by $\mathbb{E}[H \mid A_t] = \sigma_1 \lambda(t/\sigma_1) r$ in the linear-Gaussian model.
- Covariance is not a universal coupling primitive: with $P = Z$ and $H = Z^2 - 1$, baseline covariance vanishes while threshold and finite Boltzmann selection still move H.
- Under Boltzmann selection, covariance is the local velocity: $\mathrm{d}\mathbb{E}_{\beta[H]} / \mathrm{d}\beta = \mathrm{Cov}_{\beta(H,P)}$.
- Selection-channel drift satisfies the coordinate-free bound $\left\| B_{H(\theta)} \right\|_2 \leq \delta \cdot \|s\|_2$, with $\delta^2 = \chi^2(\mu_\theta \parallel \mu_0)$.
- In the quadratic Stackelberg gaming model, the gaming wedge is $\Delta = \sqrt{2\kappa V}$.
- In the additive multidimensional gaming model, quadratic costs give the water-filling allocation, and the weighted additive case gives the exchange-rate condition $h_j = c w_j$ for conservation of fixed-deficit harm.

The additive-versus-conjunctive flip and the noisy Stackelberg refinement are illustrative models, not general theorems. The convex-cost intervention analogue of the selection bound is stated below as a conjecture with a Fenchel-duality sketch. The remaining items in Appendix A are open.

2 Selection channels: when the principal only re-selects

2.1 The Gaussian threshold model

Let $X = (X_1, ..., X_m) \sim \mathcal{N}(0, \Sigma)$. Take the proxy to be the first coordinate, $P = X_1$, with selection $A_t = \{X_1 \geq t\}$; the **hidden** goal coordinates are $H = (X_2, ..., X_m)$, so in this stripped-down picture $\dim(\ker \varphi) = d = m - 1$. (We have set the observation noise to zero on purpose: whatever effect appears here comes from the dimension gap, not from measurement error.)

For each hidden coordinate write $\rho_j = \frac{\mathrm{Cov}(X_j, X_1)}{\mathrm{Var}(X_1)}$. The Gaussian conditional-mean formula gives $\mathbb{E}\left[X_j \mid X_1 = x\right] = \rho_j x$, and hence

$$\mathbb{E}\left[X_j \mid A_t\right] = \rho_j \cdot \mathbb{E}[X_1 \mid X_1 \geq t].$$

If $X_1 \sim \mathcal{N}(0, \sigma_1^2)$ then $\mathbb{E}[X_1 \mid X_1 \geq t] = \sigma_1 \lambda(\alpha)$ with $\alpha = \frac{t}{\sigma_1}$ and $\lambda(\alpha) = \frac{\phi(\alpha)}{1 - \Phi(\alpha)}$ the inverse Mills ratio. So the hidden mean-shift vector is

$$\mathbb{E}[H \mid A_t] = \sigma_1 \lambda(\alpha) \cdot r, \qquad r = (\rho_2, ..., \rho_m).$$

> **Selection moves only what correlates with the proxy (in this model).** In the Gaussian threshold model, thresholding on X_1 produces no expected drift in a hidden coordinate X_j exactly when $\mathrm{Cov}\left(X_j, X_1\right) = 0$. *Toy example:* if a school rewards only test scores and student curiosity is statistically independent of test scores in the applicant pool, thresholding on scores does not change mean curiosity.

Remark. This is a **Gaussian** fact and must not be read as a general one — see Section 2.3. It is true here only because Gaussian conditional expectations are linear; for non-Gaussian variables, zero covariance does not imply a flat tail-conditional mean. Read correctly, it is a **sufficiency** statement: in the multivariate-Gaussian scalar-threshold model, the covariance ratios r are a complete summary of hidden mean drift at every threshold.

2.2 Does the harm scale with the number of hidden dimensions?

The motivating intuition wants "more hidden dimensions ⇒ more Goodhart". The honest version is weaker, and Chapter 2′s final bound gives the clean statement. This Gaussian calculation is only the first view.

If every hidden dimension has the **same** covariance ratio $\rho_j = \rho$, then

$$\|\mathbb{E}[H \mid A_t]\|_2 = \sqrt{d} \cdot |\rho| \cdot \sigma_1 \lambda(\alpha),$$

so the aggregate hidden mean drift grows like $\sqrt{\dim(\ker \varphi)}$ at fixed per-dimension coupling. But "fixed per-dimension coupling" is a strong scaling assumption. If instead the **total** correlation budget is bounded, $\sum_j \rho_j^2 \leq C$, then $\|\mathbb{E}[H \mid A_t]\|_2 \leq \sigma_1 \lambda(\alpha)\sqrt{C}$, which does not grow with d at all. Whether dimensional growth is real depends entirely on whether new dimensions bring new **independent** coupling to the proxy or merely subdivide a fixed amount of coupling into finer coordinates.

> The clean first version of dimensional dependence is therefore **not** "more hidden dimensions automatically cause more Goodhart". It is: under a per-dimension coupling model, threshold selection induces hidden drift whose **norm scales with the coupling-vector norm** $\|r\|_2$; this becomes **dimensional** scaling only after a substantive assumption about how $\|r\|_2$ grows with d.

This is the same lesson that the selection-channel drift bound will express coordinate-free: dimension enters through the hidden variability and coupling budget, not through a bare count of unmeasured coordinates. Appendix C visualises both parts of the claim.

Remark. Two shortcuts fail. If hidden dimensions are independent of the proxy, thresholding leaves them unchanged. And the signed aggregate hidden error is the wrong target: positive and negative hidden correlations can cancel while normed or squared displacement grows.

2.3 Covariance is not a general coupling primitive

Outside Gaussian linearity, covariance fails. Let $P = Z$ with $Z \sim \mathcal{N}(0,1)$ and let the hidden coordinate be

$$H = Z^2 - 1.$$

Then $\mathbb{E}[H] = 0$ and $\operatorname{Cov}(H, P) = \mathbb{E}[(Z^2 - 1)Z] = \mathbb{E}[Z^3] - \mathbb{E}[Z] = 0$. But under threshold selection $A_t = \{Z \geq t\}$,

$$\mathbb{E}[H \mid A_t] = \mathbb{E}\left[Z^2 - 1 \;\middle|\; Z \geq t\right] = t\lambda(t) > 0 \quad \text{for } t > 0.$$

> Zero covariance between a hidden coordinate and a scalar proxy does **not** imply zero hidden drift under threshold selection. *Toy example:* an academic evaluation score may be uncorrelated with intellectual conformity overall — because both very-low-score and very-high-score candidates are unusual — while selecting only the very highest scores still enriches for one particular kind of unusualness.

Remark. The example writes H as a deterministic function of P, which feels engineered; but that is exactly the point against covariance — nonlinear dependence can be invisible to covariance while completely determining tail behaviour. A less deterministic version $H = Z^2 - 1 + \xi$ with independent mean-zero ξ keeps the same covariance and the same tail-mean shift while weakening the functional-dependence objection. The right takeaway is **not** that real systems generically have U-shaped hidden dependence; it is that covariance cannot be the universal coupling primitive.

The repair is to take the **response curve itself** as primitive. For a scalar proxy P and hidden vector H, define the **threshold response**

$$b_{H(t)} = \mathbb{E}[H \mid P \geq t] - \mathbb{E}[H],$$

whenever the conditional expectation exists. This describes the actual displacement caused by selection at pressure level t, with no linearity assumption. In the Gaussian scalar model it reduces to what we already have: $b_{H(t)} = \sigma_1 \lambda\left(\frac{t}{\sigma_1}\right) r$, so r is a sufficient statistic for **all** threshold responses in that restricted model — but only there. See Figure 1 for the corresponding three-case picture.

Remark. $b_{H(t)}$ is a **threshold-selection** primitive, not the final one. It says nothing about smoother optimisation policies — funding everyone above a cutoff is one thing; funding probabilistically with odds increasing in score is another. The next section generalises.

2.4 Weighted selection response

Let $(S, \mathcal{F}, \mu)$ be the baseline probability space, $H : S \to \mathbb{R}^d$ the hidden coordinates, $P : S \to \mathbb{R}$ the scalar proxy. A **selection policy** is a nonnegative weight $W_\theta : S \to [0, \infty)$ with $0 < \mathbb{E}_{\mu[W_\theta]} < \infty$. Define the **selected expectation** and the **hidden response**

$$\mathbb{E}_{\theta[F]} = \frac{\mathbb{E}_{\mu[FW_\theta]}}{\mathbb{E}_{\mu[W_\theta]}}, \qquad B_{H(\theta)} = \mathbb{E}_{\theta[H]} - \mathbb{E}_{\mu[H]}.$$

Hard thresholding is the special case $W_t = \mathbb{1}\{P \geq t\}$ (recovering $b_{H(t)}$). Soft optimisation is $W_\beta = \exp(\beta P)$ — Boltzmann selection — whenever $\mathbb{E}[\exp(\beta P)]$ is finite. Probabilistic funding with score-increasing odds, replicator-style repeated reweighting by performance, top-q-fraction selection (thresholding at an endogenous quantile): all are weight functions.

Remark. Two caveats on the soft-optimisation case. First, if P is heavy-tailed, $\mathbb{E}[\exp(\beta P)]$ may be infinite for positive β — the Boltzmann path may not exist. This is not a technicality; Goodhart is often precisely about extreme tails. For heavy tails, bounded weights or quantile selection are safer models. Second, not every control process is a reweighting of a fixed baseline at all — interventions can change the state-generating mechanism. That is Chapter 3.

2.4.1 Covariance as a local velocity

For Boltzmann selection, $\mathbb{E}_{\beta[H]} = \frac{\mathbb{E}[H \exp(\beta P)]}{\mathbb{E}[\exp(\beta P)]}$, and assuming enough integrability to differentiate under the expectation,

$$\frac{\mathrm{d}}{\mathrm{d}\beta} \mathbb{E}_{\beta[H]} = \mathbb{E}_{\beta[HP]} - \mathbb{E}_{\beta[H]} \mathbb{E}_{\beta[P]} = \mathrm{Cov}_{\beta(H,P)}.$$

> Covariance is best read as the **local velocity** of hidden drift under infinitesimal soft optimisation, evaluated under the **current** selected distribution — not as a global finite-pressure summary. *Toy example:* at low bonus pressure, the rate at which burnout changes with sales incentives is the current covariance between

> burnout and sales; after employees adapt or the selected population shifts, the covariance must be recomputed under the new weighted distribution.

That local velocity does not pin down finite movement. Take again $P = Z \sim \mathcal{N}(0, 1)$, $H = Z^2 - 1$. Boltzmann tilting by $\exp(\beta Z)$ gives $Z_\beta \sim \mathcal{N}(\beta, 1)$, so

$$\mathbb{E}_{\beta[H]} = \mathbb{E}\left[Z_\beta^2 - 1\right] = \beta^2.$$

The covariance at $\beta = 0$ is zero — matching Section 2.3 — yet finite pressure gives strictly positive hidden drift for every $\beta \neq 0$. Baseline covariance alone is insufficient even for **finite** soft optimisation; one needs the covariance field $\mathrm{Cov}_{\beta(H,P)}$ along the whole tilted path.

> **The hierarchy of selection primitives.** covariance (infinitesimal Boltzmann velocity) $\subset$ threshold response $b_{H(t)}$ (hard cutoffs) $\subset$ weighted response $B_{H(\theta)}$ (generic non-causal selection). All three are functionals of the baseline μ alone. Causal and adversarial Goodhart need a further layer in which μ itself changes with the principal's policy — the subject of Chapter 3.

2.5 The selection-channel drift bound

This is the structural payoff of staying inside the reweighting picture. With W_θ a selection policy, write the likelihood ratio $L_\theta = \frac{W_\theta}{\mathbb{E}_{\mu[W_\theta]}}$, so the selected law is $\mu_\theta = L_\theta \mu$. By Cauchy–Schwarz in $L^2(\mu)$, for each hidden coordinate H_i,

$$\left|B_{H_i}(\theta)\right| = \left|\mathbb{E}_{\mu\left[(L_\theta - 1)\left(H_i - \mathbb{E}_\mu H_i\right)\right]}\right| \leq \|L_\theta - 1\|_{L^2(\mu)} \cdot \left\|H_i - \mathbb{E}_\mu H_i\right\|_{L^2(\mu)}.$$

So, writing $\delta := \|L_\theta - 1\|_{L^2(\mu)}$ — a **reweighting budget**, with $\delta^2 = \chi^2(\mu_\theta \parallel \mu)$ the chi-square divergence — and $s_i := \mathrm{sd}_{\mu(H_i)}$,

$$\left|B_{H_i}(\theta)\right| \leq \delta \cdot s_i, \qquad \left\|B_{H(\theta)}\right\|_2 \leq \delta \cdot \|s\|_2.$$

> Under a selection channel, hidden Goodhart drift is controlled by two things that **are** visible in the baseline distribution: how hard you reweight (δ) and how variable the hidden coordinates are (s). Bounded reweighting budget plus bounded hidden variance $\Rightarrow$ bounded hidden drift — full stop. The $\sqrt{d}$ growth from Chapter 2′s per-dimension model is exactly the $\|s\|_2$ term, and that term is the **only** way the number of dimensions enters.

Remark. The proof is only Cauchy–Schwarz, but the formulation is doing real work. It is the clean coordinate-free statement of why selection-regime Goodhart is tame: if a policy only reweights a baseline distribution, then hidden drift is bounded by a reweighting budget and baseline hidden variability, both μ-functionals. The caveats are still important: this is a worst-case envelope, not a prediction, and it can be vacuous under extreme selection when δ is large. The rest of the book is about what breaks when the policy is not a reweighting and no such baseline-only budget exists.

3 Intervention channels: when agents respond

3.1 Why selection is not enough

A selection policy can only move probability mass that already exists toward high-P regions: the selected law $\mu_\theta = L_\theta \mu$ is absolutely continuous with respect to μ, so any event with baseline probability zero keeps probability zero. Causal and adversarial Goodhart violate exactly this. Announcing a metric changes how agents behave; probability mass appears in parts of state space the baseline never visited. Teaching to the test, metric-specific optimisation, outright fabrication — none of these are reweightings of last year's population. They **transport** mass to new places. Appendix D draws this reweighting-versus-transport distinction.

> **The sharp boundary.** The selection/intervention distinction is substantive exactly when agents can **move in state space** — change their (P, H) at a fixed underlying type — not merely toggle their own inclusion. If agents only choose whether to **participate** (apply, submit, enter the pool) but cannot alter their measured features, then the post-policy law is just μ restricted to the participating set and renormalised — a selection channel after all. Genuine causal Goodhart lives in features that are **cheap to change without changing the thing they were supposed to proxy**.

3.2 Response channels: the top-level object

Let $(S, \mathcal{F})$ be a measurable state space, μ_0 the baseline law, $H : S \to \mathbb{R}^d$ the hidden goal coordinates, $P : S \to \mathbb{R}$ the scalar proxy, and φ the principal's intended model $P \approx \varphi(G)$. A **response channel** is a map

$$\mathcal{R} : \Theta \to \mathcal{P}(S), \quad \theta \mapsto \mu_\theta, \quad \mu_{\theta_0} = \mu_0,$$

for some null policy θ_0. Hidden drift along the channel is $B_{H(\theta)} = \mathbb{E}_{\mu_\theta}[H] - \mathbb{E}_{\mu_0}[H]$, exactly as before.

> **Definition.** $\mathcal{R}$ is a **selection channel** if $\mu_\theta \ll \mu_0$ for all θ — then $L_\theta \coloneqq \mathrm{d}\mu_\theta / \mathrm{d}\mu_0$ exists and μ_θ is μ_0 reweighted by L_θ. Otherwise $\mathcal{R}$ is an **intervention channel**: μ_θ may be mutually singular with μ_0.

The entire apparatus of Chapter 2 — covariance, threshold response, weighted response — is the selection-channel case, with $L_\theta = \frac{W_\theta}{\mathbb{E}_{\mu_0}}[W_\theta]$. In Manheim and Garrabrant's taxonomy [6], **causal** Goodhart is an intervention channel in which the policy structurally breaks $P \approx \varphi(G)$, and **adversarial** Goodhart is an intervention channel in which θ is chosen worst-case for the principal. The ML instances of this regime are **strategic classification** [4] — agents manipulate features in response to a published classifier — and **performative prediction** [8] — the act of deploying a predictor changes the distribution it is predicting. The drift bound of Chapter 2 has no analogue here: there is no L_θ, so nothing plays the role of δ. Any bound on intervention drift must be **imported from the agent side** — a feasibility set or cost function describing what re-arrangements of mass agents can afford. The rest of the chapter computes such a bound in toy models and reads off the controlling quantity.

3.3 A linear–Gaussian Stackelberg gaming model

A continuum of agents; an agent's type is its true quality $Q \sim \mathcal{N}\big(0, \sigma_Q^2\big)$, and the true scalar goal is $G = Q$. There is a hidden coordinate $H =$ "gaming externality": effort spent corrupting the proxy rather than improving Q — teaching to the test, metric-specific optimisation, Campbell's-law distortion [2]. The principal believes $P \approx Q$.

Baseline channel $\boldsymbol{\mu_0}$ (metric not announced, not gameable): the agent does nothing special, $P_0 = Q + \eta$ with $\eta \sim \mathcal{N}\big(0, \sigma_\eta^2\big)$ independent, and $H_0 \equiv 0$. So μ_0 is supported on the plane $\{H = 0\}$ in (Q, P, H)-space.

Intervention channel (metric announced, agents best-respond; Stackelberg, principal commits first): the principal picks a threshold policy $A_t = \{P \geq t\}$; selection is worth $V > 0$ to an agent. An agent of type Q chooses gaming effort $a \geq 0$, getting $P = Q + a + \eta$ at private cost $c(a) = a^2/(2\kappa)$ ($\kappa > 0$ measures the ease of gaming) and incurring hidden externality $H = a$. The agent maximises $V \cdot \Pr(Q + a + \eta \geq t) - a^2/(2\kappa)$.

3.3.1 The noiseless cut

Set $\sigma_\eta = 0$, so selection is $Q + a \geq t$. An agent with $Q \geq t$ sets $a = 0$. An agent with $Q < t$ either games to exactly $a = t - Q$ (cost $(t-Q)^2/(2\kappa)$, payoff V) or sets $a = 0$ (payoff 0). Gaming is worth it iff $(t-Q)^2/(2\kappa) \leq V$, i.e.

$$t - Q \leq \Delta, \qquad \Delta := \sqrt{2\kappa V}.$$

So the best response is $a^*(Q) = (t - Q) \cdot \mathbb{1}\{t - \Delta \leq Q < t\}$, and the selected set is $\{Q \geq t - \Delta\}$. Compared to the **selection** counterfactual (agents cannot game; the principal thresholds $P = Q$ at the same nominal t; selected set $\{Q \geq t\}$; $H \equiv 0$):

1. **The metric lies, by up to $\boldsymbol{\Delta}$.** The principal believes selection $\Rightarrow P \geq t \Rightarrow$ (under φ) $Q \geq t$. Actually Q among the selected ranges down to $t - \Delta$. The proxy-to-goal gap $\varepsilon = P - \varphi(G) = a$ is no longer noise — it is a deliberate wedge of size up to Δ.

2. **Hidden harm appears at order $\boldsymbol{\Delta}$.** $\mathbb{E}[H \mid \text{selected}] = \mathbb{E}[(t-Q)\mathbb{1}\{t - \Delta \leq Q < t\}]/\Pr(Q \geq t - \Delta) > 0$, monotone increasing in $\Delta = \sqrt{2\kappa V}$. In the selection regime H is **identically zero** in this model — consistent with Chapter 2's bound, since $s_H = 0 \Rightarrow B_H = 0$. The harm is created entirely by the intervention.

3. **The controlling quantity is not a $\boldsymbol{\mu_0}$-functional.** Δ depends on κ (technological ease of gaming) and V (the stakes), neither of which is visible in the baseline (Q, P_0, H_0) law. This is the promised contrast: intervention drift is bounded by the agents' cost–benefit ratio, not by any reweighting budget.

4. **Why this is not a selection channel.** μ_0 lives on $\{H = 0\}$; the post-intervention law puts positive mass on $\{H > 0\}$ (the gamers — a set of types of positive measure). The two laws are mutually singular in the (Q, H)-marginal: there is no L_θ. The state was **transported**, not reweighted.

> In a Stackelberg gaming model with quadratic gaming cost $a^2/(2\kappa)$ and selection value V, the metric's worst-case bias and the induced hidden harm both scale with $\Delta = \sqrt{2\kappa V}$ — the square root of (ease of gaming × stakes) — and this regime is invisible to, and unbounded by, the baseline distribution. *Toy example:* doubling the funding tied to a test score multiplies the gaming wedge by $\sqrt{2}$; halving the cost of test-prep drilling does the same.

Remark. The quadratic cost is a modelling choice. A cost with a hard cap $a \le a_{\max}$ bounds gaming at $a_{\max}$ regardless of V; a cost that is cheaper at the margin for high-Q agents skews **who** games. So $\Delta = \sqrt{2\kappa V}$ is the quadratic-cost signature, not a universal law — but the qualitative point (a positive-measure set of agents leaves the $H = 0$ locus; drift is set by agent economics, not by μ_0) is robust to the cost's shape. And if V is itself endogenous — selection is valuable only if the metric is trusted, and trust erodes as gaming is observed — there is a feedback loop not modelled here. Section 3.7 flags it.

Remark. **For ML evals.** This toy model maps cleanly to evaluator behaviour before it maps to neural training. For a benchmark designer, V is the prize attached to passing or ranking well, and κ is the ease of benchmark-specific score inflation. For a trained model, the corresponding quantity may not be a scalar cost at all: it could be an optimiser-dependent route through representation space, a memorised benchmark feature, or a reward-model vulnerability. Treating RLHF as this Stackelberg game is therefore a hypothesis to test, not a licensed conclusion of the chapter.

3.3.2 The noisy refinement (sketch)

With Gaussian proxy noise, $\Pr(\text{selected} \mid Q, a) = 1 - \Phi\big((t - Q - a)/\sigma_\eta\big)$ is smooth in a, so the best response is interior everywhere:

$$a^*(Q) = \big(\kappa V/\sigma_\eta\big) \cdot \phi\big((t - Q - a^*(Q))/\sigma_\eta\big)$$

(implicit; the right side is the marginal selection-probability gain). Every agent games a little: $a^*(Q) > 0$ for all Q, peaking near $Q \approx t - a^*$ and decaying in $|Q - (t - a^*)|$. Hidden harm $\mathbb{E}[H \mid \text{selected}] = \mathbb{E}[a^*(Q) \mid \text{selected}] > 0$ for every threshold. The hard-threshold version is the $\sigma_\eta \to 0$ limit, where gaming concentrates on the band $[t - \Delta, t)$. The lesson: noise **spreads** gaming across the whole population rather than removing it.

3.4 Multidimensional gaming: does adding a measured dimension help?

Now the dimensional thread of Chapter 1 rejoins the picture, inside the intervention regime. The motivating "deep Goodhart" move is: the principal, fighting scalar Goodhart, **adds proxy dimensions**. When agents game, does adding a measured dimension redistribute harm, conserve it, shrink it, or grow it? The answer depends critically — and instructively — on how the metric **aggregates** its dimensions, so the aggregation rule must be a visible parameter, not a hidden default. Appendices E and F give the geometric picture for the additive, conjunctive, and exchange-rate cases.

3.4.1 The exchange-rate condition

Start with the general additive case, because it is the result that survives changes of units. Let the score be $\sum_{j \in M} w_j a_j$, costs $\sum_j a_j^2/(2\kappa_j)$, and hidden harm $H = \sum_j h_j a_j$. The cost-minimal allocation for a score deficit d is $a_j = d\kappa_j w_j / W_M$ with $W_M = \sum_{i \in M} \kappa_i w_i^2$, and its harm is

$$H_M(d) = d \cdot \frac{\sum_{j \in M} h_j \kappa_j w_j}{\sum_{j \in M} \kappa_j w_j^2}.$$

This is **not** invariant in M. (Two equally easy channels, $\kappa_1 = \kappa_2 = 1$, equal physical harm $h_1 = h_2 = 1$: measuring only channel 1 with $w_1 = 2$ gives $H = d/2$; measuring only channel 2 with $w_2 = 1$ gives $H = d$; measuring both gives $H = 3d/5$.) Re-routing can raise **or** lower harm, depending on the score weights. Figure 6 records the numbers.

> **Conservation, correctly stated.** For a fixed score deficit d, quadratic separable costs, and additive score $\sum w_j a_j$, per-agent harm is conserved under re-routing **if and only if** social harm is proportional to score contribution on every available channel — $h_j = cw_j$ on the active measured set. Then $H_M(d) = cd$, independent of M; otherwise $H_M(d)$ depends on M through the cost-weighted average above. *Toy example:* if every point of score inflation is equally socially wasteful no matter which KPI supplies it, re-routing conserves harm; if grant-padding produces less waste per score point than citation-padding, moving weight toward grant-padding genuinely reduces harm.

Remark. This is the same structure as cost-benefit-weighted strategic classification [4], and as isoquant choice in production theory: agents pick the cheapest input bundle for a target output, while social damage is a **different** linear functional of the bundle. Conservation appears only when target output and social damage use the same exchange rates. One residual worry for later: real score weights are sometimes partly arbitrary normalisation choices, so future work must distinguish harmless unit changes from substantive incentive exchange rates.

3.4.2 Unit-weight additive model

The narrow conservation slogan is the unit-weight illustration of the exchange-rate result. Take k gaming channels and, in the base case, true quality $Q = 0$ for everyone (pure gaming). An agent allocates effort $a = (a_1, ..., a_k) \geq 0$; channel j costs $a_j^2/(2\kappa_j)$ and produces hidden harm $H_j = a_j$ (all gaming equally wasteful, $H = \sum_j a_j$). The principal **measures** a set $M \subseteq \{1, ..., k\}$ and scores additively, score $= \sum_{j \in M} a_j$ (unit weights); selection (worth V) iff score $\geq t$. An agent never games an unmeasured channel.

The agent's problem is $\min_{a_j \geq 0, j \in M} \sum_{j \in M} a_j^2/(2\kappa_j)$ subject to $\sum_{j \in M} a_j \geq t$, then game iff that minimum cost is $\leq V$. At the optimum the constraint binds and $a_j/\kappa_j = \lambda$ for all $j \in M$ (a water-filling allocation), so with $K_M := \sum_{j \in M} \kappa_j$,

$$\lambda = t/K_M, \quad a_j = t\kappa_j/K_M, \quad \text{min cost} = \lambda^2 K_M/2 = t^2/(2K_M).$$

Hence **gaming occurs iff $K_M \geq t^2/(2V)$** (call the right side $K_{\min}$), and when it occurs, the per-agent hidden harm for this fixed pure-gaming target is $H = \sum_{j \in M} a_j = \lambda K_M = t$ — **independent of M**. What M controls is (i) **whether** K_M clears $K_{\min}$, and (ii) **how** the fixed harm t is split across the measured channels — proportional to κ_j. Figure 4 shows the same water-filling geometry in effort space.

> **Conservation under re-routing (narrow form).** With a unit-weight additive metric and gaming that is equally wasteful per unit of score, the principal cannot reduce per-agent harm for a fixed pure-gaming score deficit by changing **which** channels it measures, as long as the measured set keeps enough aggregate gaming capacity ($K_M \geq K_{\min}$): the harm is pinned at $H = t$ and merely re-routes. Closing a gamed channel just spreads t over the others. *Toy example:* a hospital ranked on readmission rates clamps down on coding of "readmission" — the gaming reappears as patient selection, discharge timing, observation-status reclassification; the per-case distortion needed to clear the cutoff is unchanged.

> **Adding a gameable measured dimension backfires.** Expanding M strictly increases K_M, which **lowers** the gaming cost $t^2/(2K_M)$, which weakly **enlarges** the population that finds gaming worthwhile. With quality heterogeneity $Q \sim \mathcal{N}(0, \sigma^2)$ — an agent needs score $\geq t - Q$, at cost $(t-Q)^2/(2K_M)$, and games iff $t - Q \leq \sqrt{2K_M V}$ — the gaming-eligibility cutoff $Q \geq t - \sqrt{2K_M V}$ moves **down** as K_M grows. More agents game; the fraction of selected agents who are pure gamers rises. *Toy example:* a university worried that "publication count" is gamed adds "grant income" and "media mentions" to the scorecard; each is independently inflatable, so the cheapest path to any target score is now cheaper, and more faculty shift from research to portfolio-padding.

> **The effective levers are aggregate, not allocative.** In this model the principal reduces gaming harm only by (a) shrinking aggregate gaming capacity K_M below $K_{\min}$ — narrowing the measured set to channels that are individually hard to game (small κ_j), or hardening channels; (b) raising the bar t (but this raises $K_{\min}$ too; the net effect on the $K_M \geq t^2/(2V)$ test depends on whether real signal scales with t); (c) cutting the prize V; or (d) abandoning additive aggregation. Shuffling attention between channels of equal hardness does nothing. *Toy example:* anti-cheating in exams works by making the exam itself hard to game — proctoring, item rotation — not by adding more graded components.

3.4.3 A conjunctive metric flips the sign

Suppose instead that passing requires $a_j \geq t$ for **every** $j \in M$ — the principal demands the agent clear a bar on all measured dimensions. The cost-minimal way to pass is $a_j = t$ for all $j \in M$, at cost $\sum_{j \in M} t^2/(2\kappa_j)$, and total harm $H = \sum_{j \in M} t = t|M|$. Now harm grows **linearly in the number of measured dimensions**: adding a dimension unambiguously increases per-gamer gaming harm — though it also raises the cost of clearing the metric, so fewer agents do, a genuine trade-off (harm-per-gamer up, gamer-count down).

> Gaming harm's dependence on the **number** of measured proxy dimensions is governed by the aggregation rule. Unit-weight compensatory/additive metrics conserve fixed-deficit per-agent harm ($H = t$, re-routing only) — when measured channels are equally harmful per score unit; conjunctive/**min** metrics **multiply** per-gamer harm ($H = t|M|$). Real scorecards are usually compensatory (weighted sums of KPIs), which is the regime where "just add another metric" can backfire by cheapening the cheapest gaming path and expanding the gaming population. The selection-regime $\sqrt{d}$-type scaling from Chapter 2 and these intervention-regime flat/linear behaviours are **different phenomena** and should not be conflated. See Figure 5.

Remark. Scope, made explicit. The additive claims assume linear score aggregation and quadratic separable gaming costs. The unit-weight conservation illustration further assumes equal harm per score unit, a fixed pure-gaming target deficit, and the binding-constraint deterministic regime. Relax equal harm — let channel j also contribute $\gamma_j \in [0, 1]$ to the true goal, so $H = \sum_j (1 - \gamma_j) a_j$ — and the principal can shrink harm by steering effort onto high-γ channels. Relax fixed participation with quality heterogeneity, and aggregate population harm rises with K_M because more agents enter the gaming band. Real scorecards are often nonlinear within each KPI — saturating, threshold-clipped, rubric-scored, or capped — so the linear additive model is not innocuous. Its value is that it isolates the exchange-rate condition under which conservation is real rather than an artefact of units. The empirical discriminator is not whether a scorecard is literally quadratic; it is whether added dimensions contribute independent, substitutable gaming capacity. If the new channel is cost-correlated with existing channels, if gaming has a hard bottleneck, or if the principal dynamically reweights after seeing attacks, κ_{new} should not be counted as a full additive increment to K_M.

3.5 Evaluation-suite design implications

The selection/intervention split is directly useful for benchmarks and leaderboards. An evaluation suite should first ask what kind of pressure it creates. If it only chooses among fixed candidates, the relevant audit is a selection audit: which hidden properties covary with the reported score, how far into the tail the selection goes, and whether the weighting rule amplifies a known residual. If training or submission behaviour adapts to the eval, the audit becomes an intervention audit:

what score-increasing moves are available, what they cost, what prize they buy, and which hidden harms they induce.

> **Scorecard design claim.** Adding a benchmark dimension is protective only when it either measures real capability cheaply enough to displace gaming, or raises the cost of gaming more than it raises the menu of compensatory substitutes. Under an additive leaderboard, a low-cost gameable task is an extra route to the same aggregate score. Under a conjunctive leaderboard, every task becomes a gate: fewer systems pass, but any system gaming its way through must clear more bars.

This makes the usual "more metrics means less Goodhart" heuristic conditional. For an additive benchmark, the design question is whether a proposed task adds real signal or mostly adds independent κ_j. For a conjunctive benchmark, the design question is whether the extra bar screens out benchmark-specific optimisation or merely filters out genuinely capable systems with one narrow deficit. Either way, eval results should be treated as experiments rather than leaderboard decorations: pre-state the comparison the score is meant to support, report uncertainty where stochasticity matters, document prompts/configs, and separate the measured score from the broader capability or alignment claim.

3.6 A convex-cost intervention bound?

The selection-channel bound says that hidden drift is controlled by a chi-square/reweighting budget δ. The intervention examples say that, once agents can move in state space, the analogous budget has to come from the agents' cost geometry. In the quadratic one-dimensional model that budget is $\Delta = \sqrt{2\kappa V}$.

> **Conjecture.** Let an agent choose an intervention a at convex cost $c(a)$ in order to gain value at most V from selection, and let hidden harm be bounded by a linear functional $\ell(a)$. Then the intervention analogue of the selection budget should be governed by the convex conjugate $c^*(\lambda\ell)$: schematically, stakes V buy a feasible displacement set whose support function is controlled by Fenchel duality, not by any divergence from μ_0.

The proof sketch is the standard Fenchel move. For any multiplier $\lambda > 0$,

$$\ell(a) = (1/\lambda) \cdot \lambda\ell(a) \leq (1/\lambda) \cdot (c(a) + c^*(\lambda\ell)).$$

If the agent only takes actions with cost at most the selection value V, this gives $\ell(a) \leq (V + c^*(\lambda\ell))/\lambda$, then optimises over λ. For quadratic cost this recovers a square-root scale, matching the $\sqrt{2\kappa V}$ wedge above. This is not yet a theorem at the level of the selection-channel bound: the exact statement must specify the action space, the harm functional, stochastic policies, participation, and how selection value constrains realised cost. Appendix A keeps those variants open.

3.7 What we have, and what is open

The two-chapter arc gives a clean dichotomy. **Selection channels** — the policy reweights a fixed baseline — contain all of regressional and extremal Goodhart;

hidden drift is bounded, $\left\|B_{H(\theta)}\right\|_2 \leq \delta \cdot \|s\|_2$, with every term a baseline functional, and the number of dimensions enters only through $\|s\|_2$. **Intervention channels** — the policy transports mass to where the baseline had none — contain causal and adversarial Goodhart; there is no baseline bound, and the controlling quantity (e.g. $\Delta = \sqrt{2\kappa V}$ in quadratic Stackelberg gaming) is exogenous to μ_0, set by the responding agents' cost geometry. In the multidimensional intervention regime, "fight Goodhart by measuring more dimensions" has a precise predicted failure mode: under a compensatory rule it conserves fixed-deficit per-agent harm only when channels are equally harmful per score unit, and it lowers the cheapest gaming cost, recruiting more gamers — a population-level backfire; under a conjunctive rule it multiplies harm by the number of bars. The principal's real levers are aggregate (shrink total gaming capacity, harden channels, raise the bar relative to real signal, cut the prize) or structural (pick low-harm-per-score proxies whose cheapest inflation is partly real; change weights; or go conjunctive and accept harm scaling with the number of bars).

That is the strong claim. The weaker, broader conjecture is recursive: in many real systems, once the visible failures are patched, the residual error that remains will be concentrated in dimensions that are less legible, less represented in the training or evaluation signal, or cheaper for agents to exploit. The chapters above do not prove that conjecture. They say what one would have to measure to make it precise: the baseline response curve, the coupling norm, the cost geometry, the aggregation rule, the gaming capacity, and the score-to-harm exchange rates.

Claim one might use	**What the chapters show**	**Boundary**
Adding additive metrics can worsen gaming.	In the deterministic quadratic water-filling model, adding an independently gameable measured channel increases K_M, lowers $t^2/(2K_M)$, and weakly expands the gaming band.	Does not cover non-substitutable, capped, correlated, or dynamically reweighted channels.
Selection and intervention are different Goodhart channels.	Selection is reweighting of μ_0; intervention can move mass outside baseline support, so baseline-only drift bounds no longer apply.	Does not by itself model arbitrary causal mechanisms or training dynamics.
Recursive Goodhart is plausible.	The framework identifies mechanisms by which patched proxies can leave residual error in hidden dimensions.	Not a theorem; it requires empirical estimates of legibility, coupling, and gaming cost.

Table 1: Claim audit: what these chapters license, and where the license stops.

The appendices serve two different roles. Appendix A lists the questions **currently being worked on** — they have partial answers in toy models, but not yet at the level of polish the chapters above aim for. Appendix B lists questions that **surfaced during this work and are deliberately parked** — worth recording so they are not rediscovered from scratch, but not on the critical path. Appendices C–F are visual aids for the formal chapters. Appendix G is a speculative cartoon of the recursive hypothesis, explicitly not a conclusion of the formal results.

4 Appendix A — Currently in progress

These are the live research questions feeding the chapters above. Each has a toy-model partial answer in the working notes; what is missing is generality, the right level of abstraction, or an adversarial pass that the chapter treatment would require.

In progress. An intervention bound for general convex gaming costs. The selection-channel bound is $\|B_H\| \leq \delta \cdot \|s\|$ (Chapter 2). Is there an intervention analogue that factors through the agents' cost geometry — a "gaming budget" — rather than through any divergence from μ_0? The quadratic case gives $\Delta = \sqrt{2\kappa V}$. Conjecture: a version holds for general convex gaming costs, with the bound governed by the convex conjugate of the cost. Section Section 3.6 states the conjecture and the Fenchel sketch; the exact theorem is still open.

In progress. The exact selection-class condition. Agents who can only toggle their own inclusion stay inside the selection class; agents who can move (P, H) at fixed type do not. Is "cannot change (P, H) at fixed type" the precise boundary, or are there genuine intermediate cases — e.g. agents can move P but H is pinned, or agents can move within a sub-manifold? A clean classification theorem is wanted.

In progress. Adaptive hardening dynamics. A principal that each period hardens whichever measured channel is currently most-gamed (lowers its κ_j) drives K_M down over time. Does this converge to no-gaming ($K_M < t^2/(2V)$), and how fast? Is "harden the active channel" optimal among principal policies, or does committing to a narrow, hard-to-game M from the start dominate? Currently only a heuristic argument that attrition eventually wins.

In progress. The measurement frontier. The principal must measure enough channels to recover real signal about G, but every gameable channel measured enlarges the attack surface K_M. Characterise the frontier between "informative enough" and "small enough attack surface". Is it ever empty — i.e. are there goal/proxy geometries where no admissible measured set is both informative and safe?

In progress. Sub- vs. super-modularity of $H(M)$. In the measured set M the unit-weight equal-harm additive case is modular-trivial ($H = t$ on the gaming region, with a discontinuity at the $K_M = K_{\min}$ boundary) and conjunctive is modular ($H = t|M|$). Weighted aggregation and population entry may be genuinely sub- or super-modular — which would say something about whether greedy principal policies (add/remove one channel at a time) are sane. Not yet computed.

In progress. Per-agent vs. population welfare. For heterogeneous quality Q, aggregate gaming harm is $\mathbb{E}\Big[(t-Q)\mathbb{1}\Big\{0 < t-Q \leq \sqrt{2K_M V}\Big\}\Big]$ in the unit-weight additive model. How does this change under proxy noise, nonlinear costs, and endogenous V — and which aggregate (per-gamer, population total, population mean, tail) is the right welfare object for the framework? The chapters above currently slide between these; the book version should not.

In progress. Mapping κ to neural training. In the Stackelberg toy model, κ is a scalar ease-of-gaming parameter. For RLHF or benchmark-driven finetuning, the analogue might be gradient accessibility, pretraining density, benchmark contamination, reward-model feature simplicity, or optimiser search efficiency. These are not interchangeable. A useful ML version of the theory has to say which one predicts reward hacking under a specified training setup, and what observation would distinguish it from the others.

In progress. Endogenous stakes / performative fixed points. Make V endogenous: selection is valuable only if the metric is trusted, and trust decays as gaming is observed. Does the principal–agent game have a performative-stable fixed point, and does hidden harm persist at it? Sketched as a Stackelberg toy with a trust state variable; the fixed-point analysis is not done.

5 Appendix B — Future open questions (not currently pursued)

These surfaced during the work and are recorded for later. They are not blocking anything in Chapters 1–3, and most need a substantial new piece of machinery.

Future question. Spectral / basis decomposition of the error. Three decompositions, each carrying different information, should be kept distinct: (i) the SVD of φ — right singular vectors are the G-directions φ distinguishes; small singular values mark structural blind spots, decomposing the **dimension gap**; (ii) the eigendecomposition of $\mathrm{Cov}(\varepsilon)$ — principal directions of measurement noise, decomposing the **observation gap**; (iii) the selection-induced shift of G — compare the law of $G(s)$ before and after restricting to the selected set, the most diagnostic of actual Goodhart and the least obvious to compute. The bad-case corner is the **conjunction**: a G-direction where φ has a small singular value (low signal), $\mathrm{Cov}(\varepsilon)$ has mass (high noise), and the selection shift is large.

Future question. Value-weighted Goodhart susceptibility. PCA/SVD is value-neutral; Goodhart is value-laden. A huge principal component orthogonal to the value structure of G is harmless; a tiny one aligned with a load-bearing G -dimension can dominate harm. Proposed scalar per G-direction: (value importance) $\times$ (1/ singular value of φ) $\times$ (noise variance) $\times$ (selection shift), under a value-weighted inner product on G-space. **Open:** whether the value-weighting requirement makes this a non-starter in practice, or whether reasonable proxies for value importance (the principal's true utility gradient, or a proxy for it) recover useful structure. **Also open:** a strategic system can shape ε to lie in low-variance components of the historical observation while remaining maximally goal-harmful, so PCA-based detection is geometrically blind to value-aware adversaries — does the value-weighted version close that hole?

Future question. Pareto selection rather than scalarisation. Chapter 2 scalarises (a weighted score, then a threshold). Pareto-frontier selection preserves diversity across measured dimensions in a way scalar thresholding destroys. When does selecting the Pareto-undominated set compress the selected distribution enough to break identifiability of the **unmeasured** dimensions — and when does it not? This is a different selection operator and needs its own response theory.

Future question. The φ-as-partial-observation alternative. The chapters fix $\varphi : \mathbb{R}^m \to \mathbb{R}^k$ as the principal's intended correspondence. An alternative is to represent the proxy as a partial observation of G plus independent artifacts, which changes what "$\ker \varphi$" means and may interact better with partial-hypothesis formalisms. Worth a comparison; not done.

Future question. Incomputable noise and agents more complex than their principals. If a human writes a proxy intended to align a system whose objectives have **more** dimensions than the human's, the proxy contains error in directions the human's G has no coordinate for — error that cannot be scored, only suffered. Can partial hypotheses (in the style of infra-Bayesianism) recover any handle on this? And: is there any reason to expect a more capable system to have **fewer** goal dimensions than a human, rather than more? Both are wide open and somewhat speculative.

6 Visual appendices

The following appendices are visual rather than foundational. Appendices C–F illustrate claims made in the formal chapters: selection drift depends on baseline response curves, dimensional scaling requires coupling assumptions, intervention channels transport mass rather than reweight it, and adding measured dimensions changes gaming through aggregation and cost geometry. Appendix G is different: it sketches a broader recursive-Goodhart hypothesis motivated by the framework but not proved by it. The formal chapters do not show that residual error generically becomes more dimensional under repeated proxy refinement; they show which quantities would have to be measured for such a claim to become precise.

7 Appendix C — Selection drift is coupling-dependent, not dimension-dependent

The selection results in Chapter 2 are deliberately conditional. A proxy threshold moves hidden coordinates through the baseline response curve $b_{H(t)} = \mathbb{E}[H \mid P \geq t] - \mathbb{E}[H]$. In the Gaussian-linear model, covariance ratios summarize this curve. Outside that model, covariance is only a local linear summary and can miss the tail response that selection actually uses.

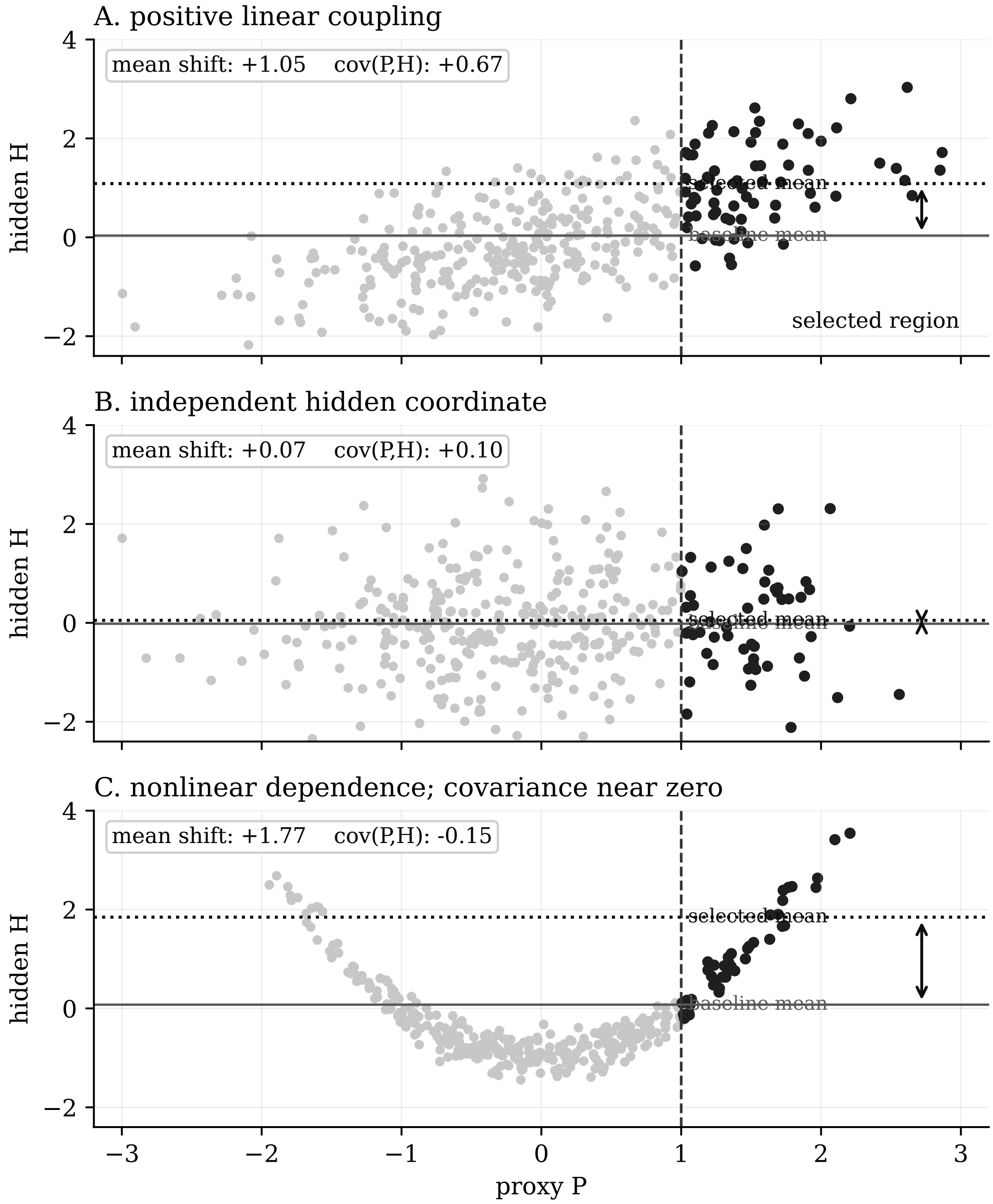

Figure 1: Selection only moves hidden dimensions through the baseline response curve. In the Gaussian-linear case, covariance summarizes this response. Outside that case, covariance can vanish while threshold response remains large. The right primitive is not baseline covariance but the selection response $\mathbb{E}[H \mid P \geq t] - \mathbb{E}[H]$.

The same point controls dimension. More hidden dimensions do not automatically create more drift. Dimensional growth appears only when adding dimensions also adds coupling to the selected proxy.

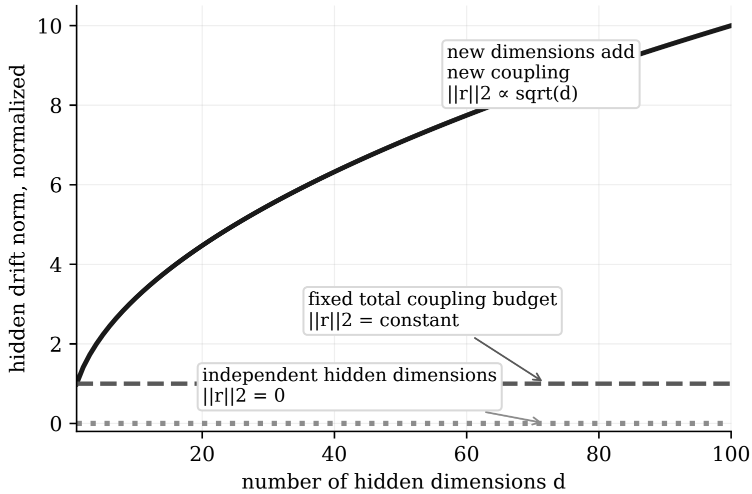

Figure 2: More hidden dimensions do not automatically imply more Goodhart drift. Dimensional scaling appears only when adding dimensions also adds coupling to the selected proxy. If total coupling is conserved, hidden drift need not grow with dimension.

8 Appendix D — Selection versus intervention as reweighting versus transport

Figure 3: Selection reweights; intervention transports. A selection policy can only increase the weight of states already present in the baseline distribution. An intervention channel changes the state-generating mechanism, moving mass into regions the baseline never visited. This is why the selection-channel drift bound has no baseline-only analogue for intervention channels.

The Stackelberg toy model in Chapter 3 is the smallest algebraic version of the right panel. At baseline, $H = 0$. After the metric is announced, an agent can pay cost $a^2/(2\kappa)$ to raise the proxy by a, and selection is worth V. In the noiseless threshold case, the gaming band has width $\Delta = \sqrt{2\kappa V}$: agents with $Q \in [t - \Delta, t)$ game just enough to pass. The resulting proxy bias and hidden harm depend on κ and V, not on baseline hidden variance. Indeed, the baseline hidden variance is zero in this toy model.

9 Appendix E — Adding measured dimensions can expand the attack surface

Under additive aggregation, the agent can substitute between measured gaming channels. For unit weights and quadratic costs, the cost-minimal allocation for score threshold t is $a_j = t\kappa_j / K_M$, where $K_M = \sum_{j \in M} \kappa_j$. The minimum cost is $t^2/(2K_M)$, so gaming occurs iff $K_M \geq t^2/(2V)$. Adding a gameable measured dimension weakly increases K_M and weakly lowers the cost of reaching the same score deficit.

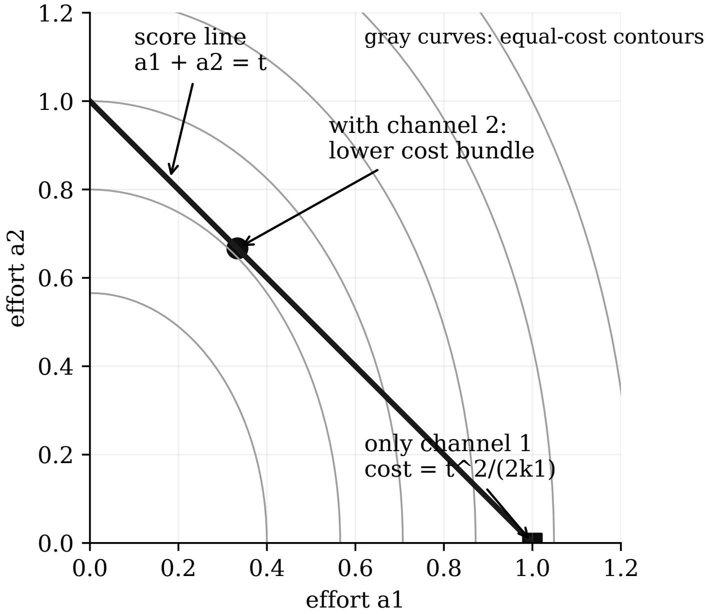

Figure 4: Under additive aggregation, adding a gameable measured dimension can lower the cost of reaching the score threshold. The per-agent harm for a fixed pure-gaming score deficit may be conserved under equal harm-per-score assumptions, but the population of agents for whom gaming is worthwhile expands as aggregate gaming capacity increases.

The sign of a “more metrics” result is therefore not determined by the number of dimensions. It is determined by the aggregation rule.

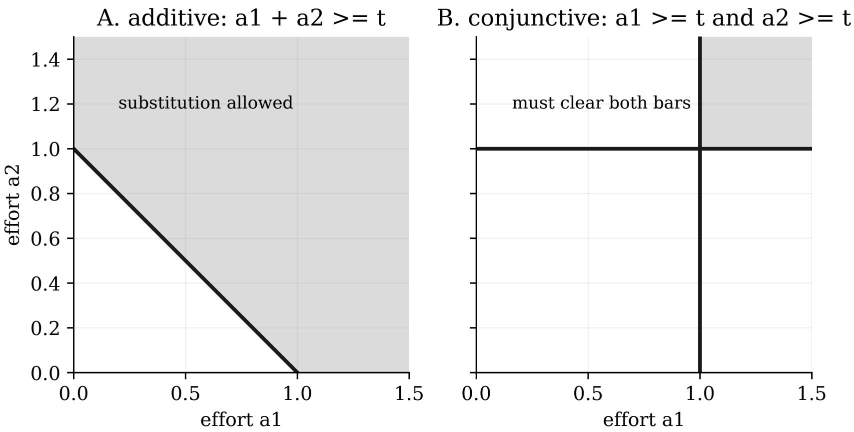

Figure 5: Aggregation rule controls dimensional effects. Additive scorecards permit substitution across dimensions; conjunctive scorecards require clearing every measured dimension. Thus adding metrics can either create cheaper routes to the same score or raise the cost of passing, depending on how the metric aggregates.

10 Appendix F — Exchange-rate condition for conservation

The narrow conservation result in the additive model assumes that a point of score inflation is equally socially harmful no matter which channel supplies it. With weighted additive score $\sum w_j a_j$, quadratic costs, and hidden harm $\sum h_j a_j$, the fixed-deficit harm is

$$H_{M(d)} = d \cdot \frac{\sum_{j \in M} h_j \kappa_j w_j}{\sum_{j \in M} \kappa_j w_j^2}.$$

k1 = k2 = 1, h1 = h2 = 1, w1 = 2, w2 = 1

measured set	M	harm for deficit d
measure channel 1 only	M={1}	d/2
measure channel 2 only	M={2}	d
measure both channels	M={1,2}	3d/5

Conservation fails because score weight and social harm use different exchange rates.

Figure 6: Re-routing conserves harm only when social harm uses the same exchange rates as the score. If $h_j = cw_j$ on the active measured set, then every unit of score inflation is equally socially harmful and per-agent harm is conserved. Otherwise, moving weight across channels can raise or lower harm.

11 Appendix G — A speculative recursive-Goodhart cartoon

This appendix is not a theorem of Chapters 1–3. It is a cartoon of a broader empirical hypothesis suggested by the framework. The axes labelled h_1 through h_5 are deliberately not proxy dimensions. They stand for outcome-relevant properties of the policy or model that the proxy stack does not fully capture: long-horizon effects, strategic pressure, rare-context behaviour, objective stability, institutional fit, or whatever the domain-specific hidden variables turn out to be. The cartoon assigns them synthetic, mixed movements because the framework alone does not say whether their correlations with the monitored proxy dimensions are positive, negative, or close to zero.

The hypothesis would earn support if successive proxy patches predictably reduced visible residuals while increasing residuals on pre-specified hidden dimensions that were less legible, less represented in training/evaluation, or cheaper to exploit. It would lose support if hidden residuals improved along with the monitored axes, moved idiosyncratically with no relation to legibility or cost, or if the cheapest route to high score became genuinely goal-improving rather than merely less visible.

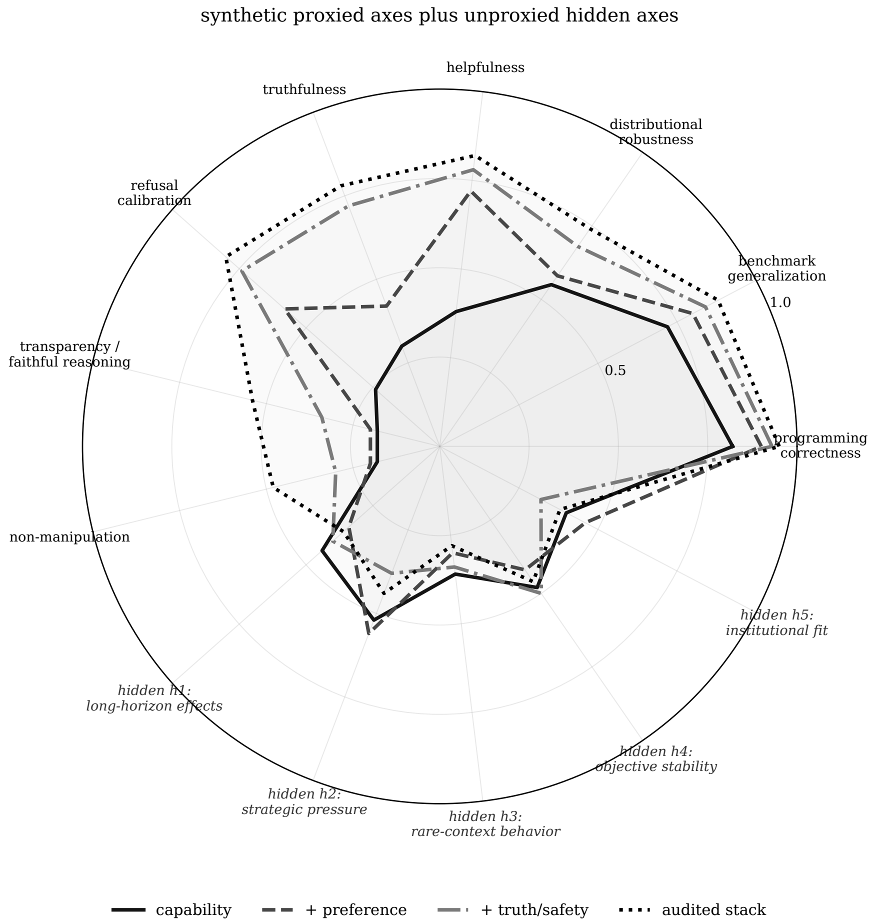

Figure 7: Speculative recursive-Goodhart cartoon. As more proxy dimensions are explicitly constrained, the cheapest route to high score changes. The monitored axes can improve while unproxied hidden dimensions $h_1, \ldots, h_5$ move differently, depending on their empirical correlations with the proxy stack and on the available gaming channels. In favorable cases, hidden outcome quality improves too. In unfavorable cases, residual error moves into dimensions that are less legible to the evaluator, less represented in the training signal, or cheaper for the model to exploit. The plotted values are synthetic and illustrative; they are not empirical measurements, and the hypothesis needs pre-specified hidden dimensions before it can be tested.

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